

# Sources of Market Power in Web Search: Evidence from a Field Experiment

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UCL

**Hunt Allcott**  
(Stanford & NBER)

**Juan Camilo Castillo**  
(Penn & NBER)

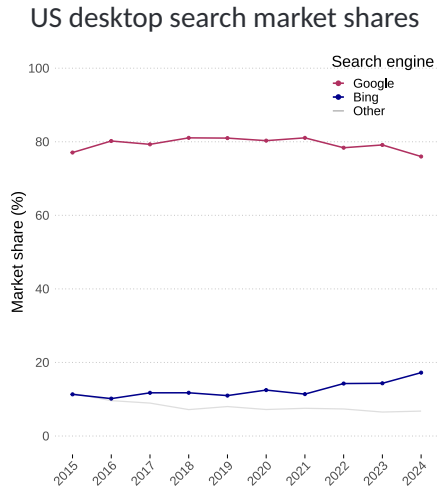
**Matthew Gentzkow**  
(Stanford & NBER)

**Leon Musolff**  
(Penn)

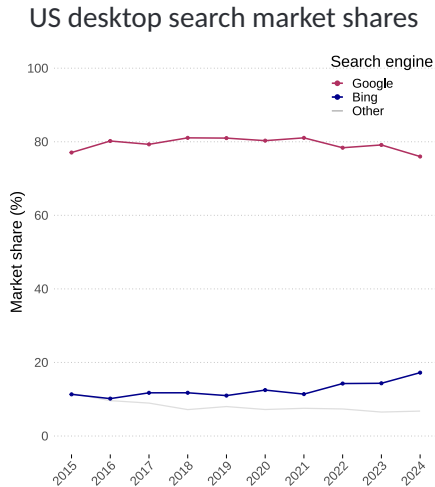
**Tobias Salz**  
(MIT & NBER)

# The web search market

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The New York Times

## *'Google Is a Monopolist,' Judge Rules in Landmark Antitrust Case*

The ruling on Google's search dominance was the first antitrust decision of the modern internet era in a case against a technology giant.



TECH

**Google paid \$26 billion in 2021 to become the default search engine on browsers and phones**

# Research questions

## 1. Why is Google's market share so high?

- True quality differences? Driven by economies of scale in data?
- Quality misperceptions?
- Default effects (switching costs/inattention)?

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- Active choice screens?
- Changing defaults?
- Requiring Google to share data with competitors?

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### **This paper:**

1. Model of demand for search engines
2. Field experiment
3. Estimate economies of scale from click-and-query data
4. Counterfactuals

# Limitations

- Desktop as opposed to mobile experiment
- Returns-to-scale analysis is narrow in scope and more speculative



# Literature

## Antitrust in web search

- UK CMA (2020), Scott Morton and Dinelli (2020), Heidhues et al. (2021), Ostrovsky (2021), Decarolis, Li, and Paternello (2023), Hovenkamp (2024)

## Competitive effect of choice frictions

- Schmalensee (1982), DellaVigna and Malmendier (2006), Handel (2013), Johnen (2019), Fowlie et al. (2021), Einav, Klopck, and Mahoney (2023), Miller, Sahni, and Strulov-Shlain (2023)

## Experience goods

- Schmalensee (1982), Shapiro (1983), Erdem and Keane (1996), Akerberg (2003), Israel (2005), Crawford and Shum (2005), Dickstein (2018)

## Returns to data

- Varian (2015), Chiou and Tucker (2017), He et al. (2017), Bajari et al. (2019), Schaefer and Sapi (2023)

## Experimental studies of digital markets

- Brynjolfsson, Collis and Eggers (2019), Allcott, Gentzkow, and Song (2020), Aridor (2022), Bursztyn et al. (2023), Blake et al. (2023), Farronato, Fradkin, and Karr (2024)

# Agenda

1. Demand Model
2. Experimental Design
3. Identification
4. Experimental Results and Estimation
5. Data Scale Economies
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$$\zeta_j^* := \underbrace{\alpha a_j^*}_{\text{Ads}} + \underbrace{\rho r_j^*}_{\text{Result Quality}} + \underbrace{\xi_j^*}_{\text{Everything else}}$$

**Quality:**

- Search result relevance, # of ads, interface, etc.

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**Payments:**

- $\eta$  is price sensitivity
- $p_{ij} = 0$  in real life, but we will pay users to switch

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**Idiosyncratic preferences:**

- Time-invariant



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Assume users unaware they misperceive quality:

- No benefits from exploration

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## Defaults:

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- Switching cost  $\sigma$  (affects marginal users)
- Inattention (affects inframarginal users)
  - Fraction  $\phi$  permanently inattentive
  - Fraction  $1 - \phi$  attentive with probability  $\pi$  each period
  - If inattentive, stick with previous choice ( $y_{it} = y_{i,t-1}$ )

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## Choice **if attentive**:

$$y_{it} = \arg \max_{j \in \{B, G\}} \left\{ E_{it}[\zeta_j^*] + \chi_{ij} - \sigma 1\{j \neq y_{i,t-1}\} \right\}$$

- No continuation value since anticipate never switching again

## Demand model

Steady-state **market share of Bing** among Chrome users:

$$(1 - \phi)F_{\Delta\chi}(\Delta\tilde{\zeta} - \sigma),$$

where  $F_{\Delta\chi}(\cdot)$  is the CDF of the differences in idiosyncratic errors and

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**Google market share can be high for four reasons:**

1. Inattention ( $\phi$ )
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... with price sensitivities governed by  $\eta$ .

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# Experiment Overview

Recruit 2,354 people on Prolific in Mar/Apr '24

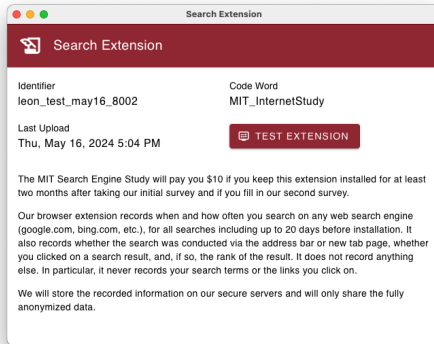
**Sample:** US adults on **desktop**

- Use only one browser: Edge or Chrome
- Usually use either Google or Bing

**Two surveys:**

- Survey 1: immediately
  - Demographics
  - Opinions about search engines
  - Install browser extension
  - Treatments
- Survey 2: 14 days later

## Search Extension



1. Records every time a search engine is used
2. Alters search result page

# Experiment Treatments

## Control (C)

- Placebo surveys

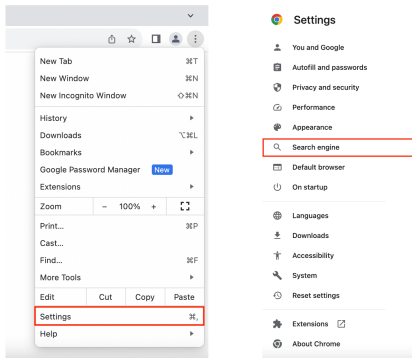
# Experiment Treatments

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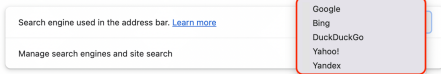
- Placebo surveys

## Active Choice (A)

- Ask for & implement preferred default



### Search engine



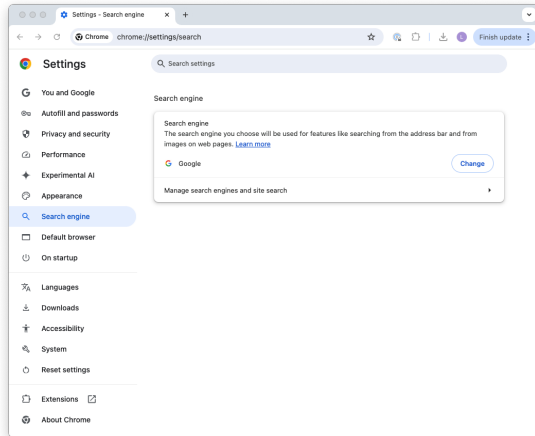
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## Switch Bonus (S)

- Offer {\$1, **\$10**, \$25} to switch for 14d
- After 14d, make active choice



# Experiment Treatments

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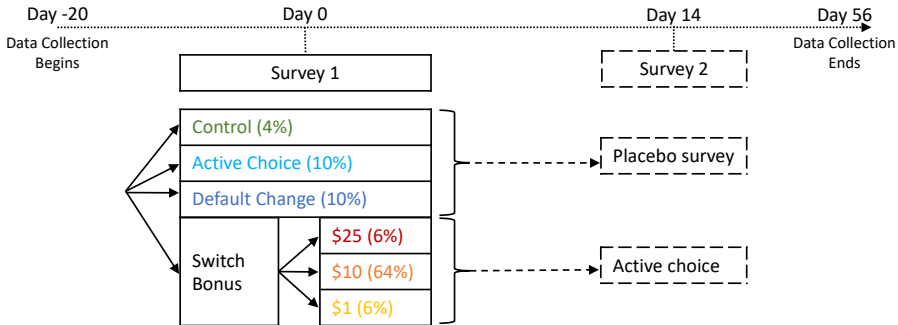
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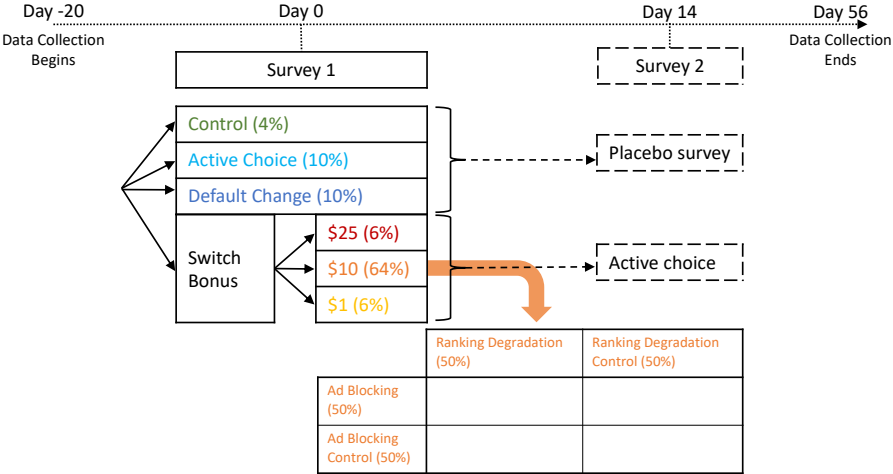
## Default Change (D)

- Offer \$10 to change default for 2d

# Details



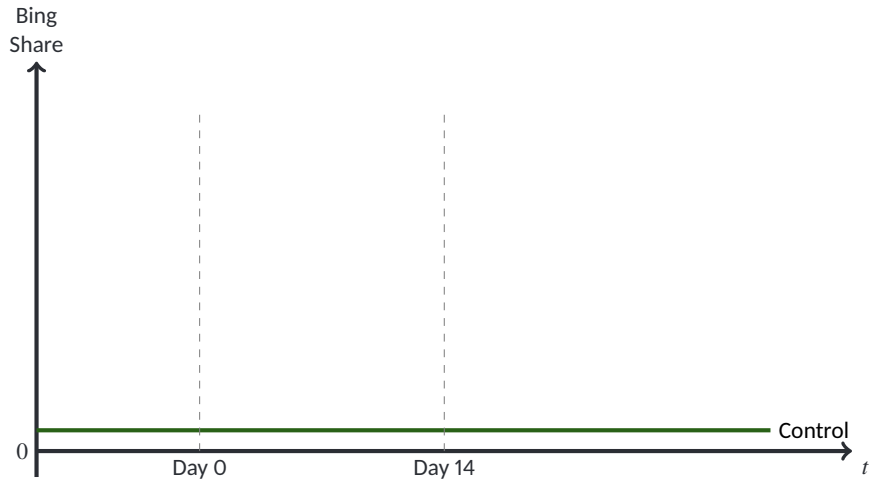
# Details



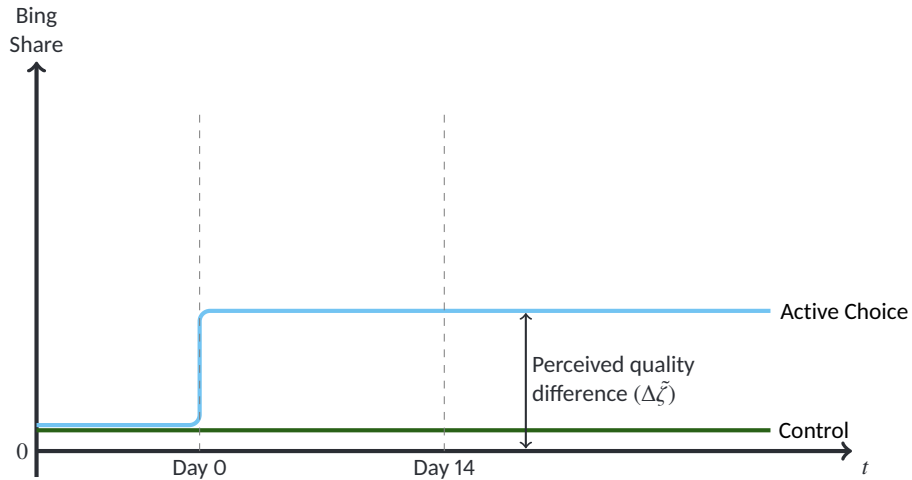
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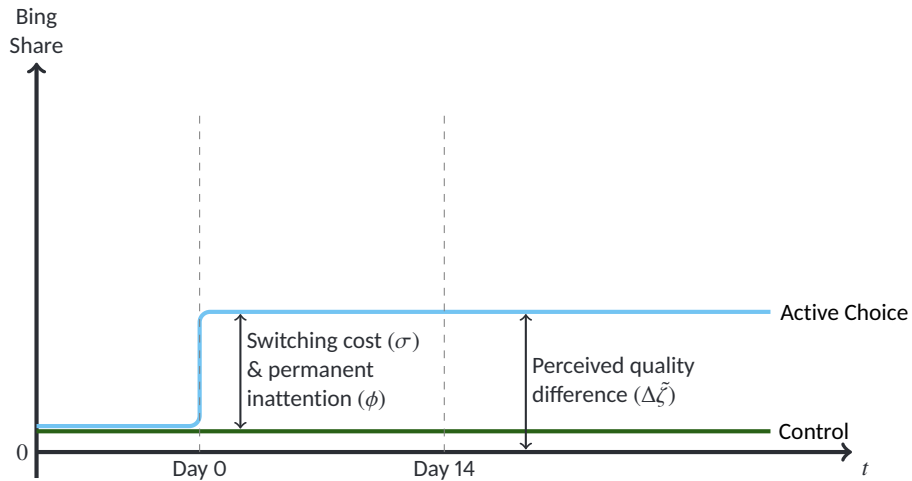
## Identification for Chrome users



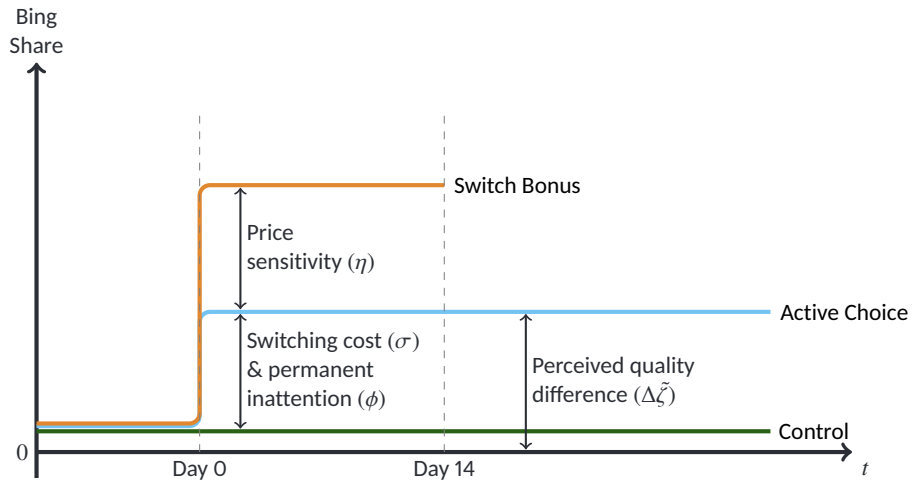
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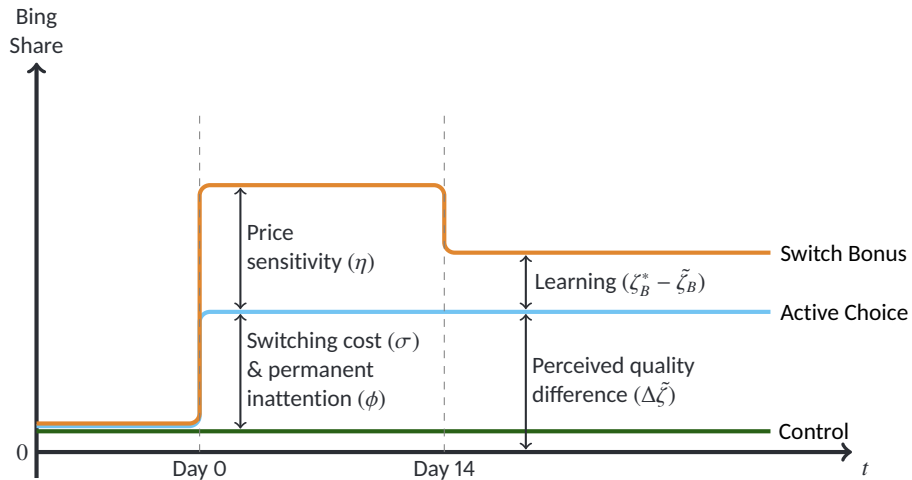


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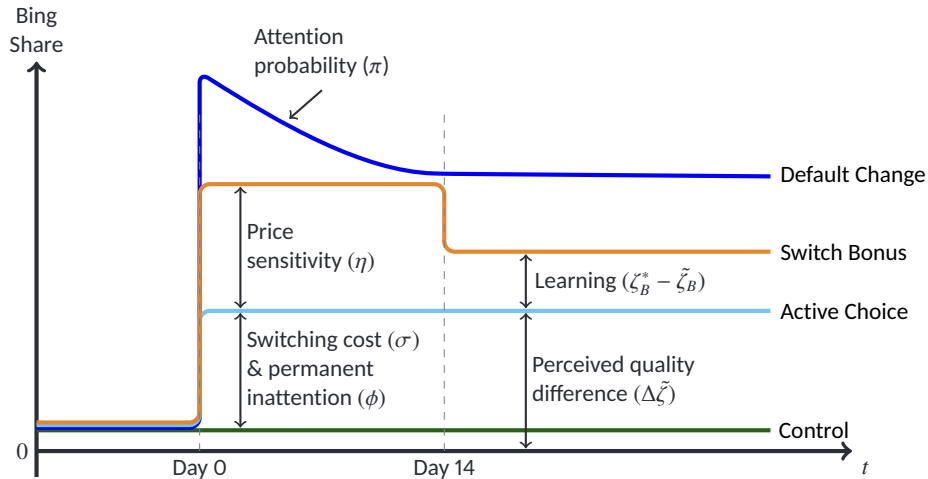




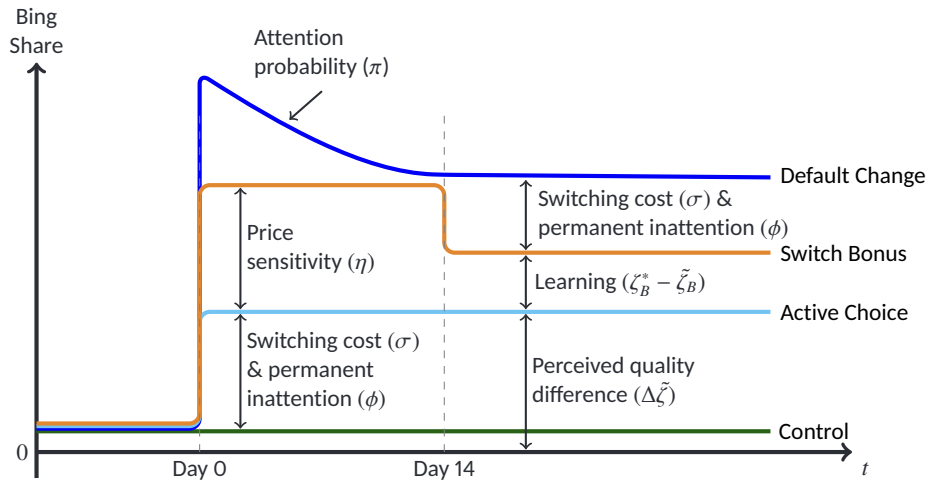
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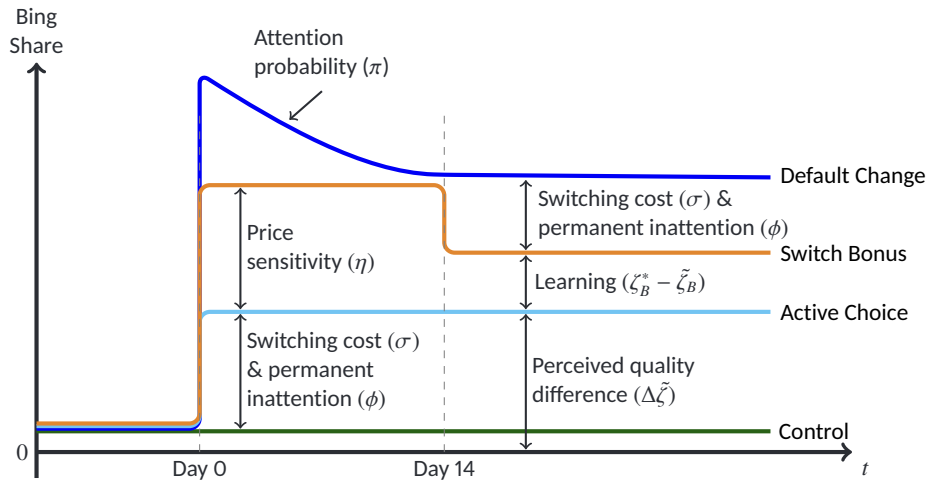
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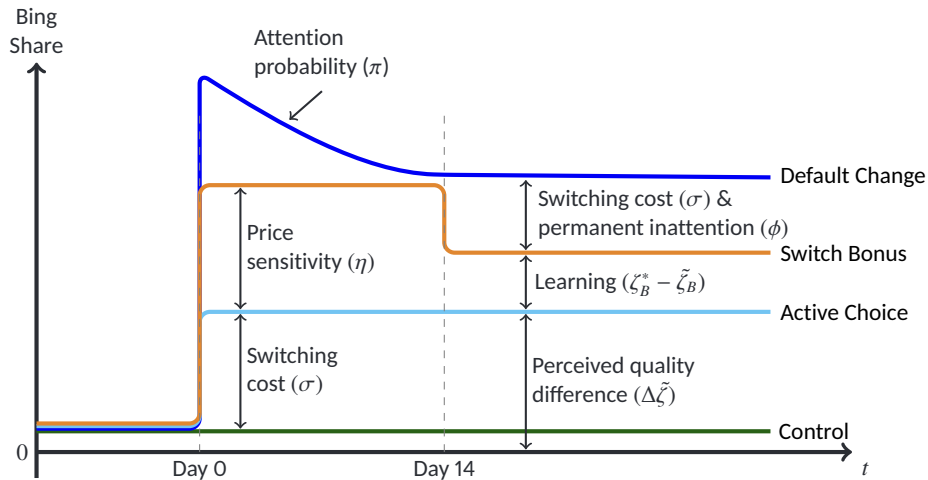


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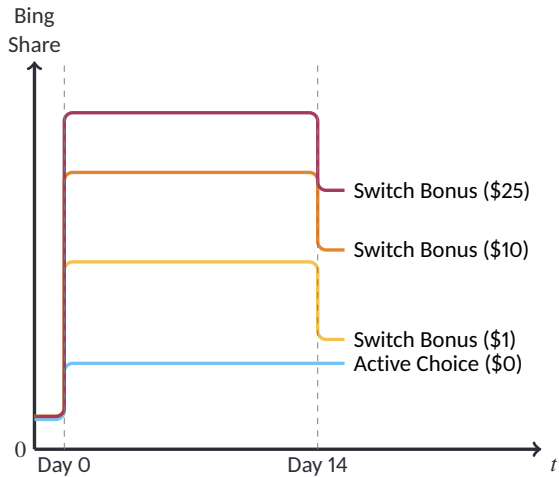
Identifying  $\sigma$  and  $\phi$ : Diff. between Active Choice and Control *almost* unaffected by  $\phi$  [More](#)

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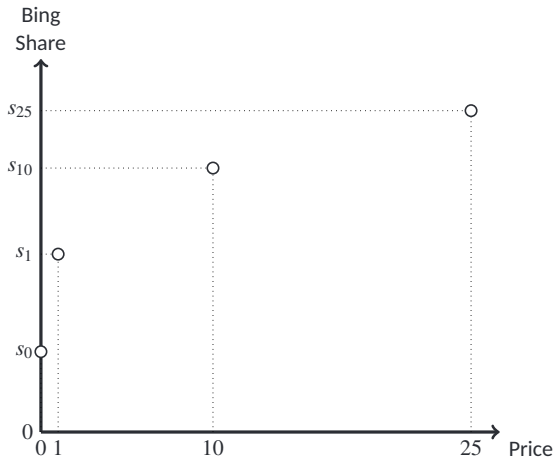
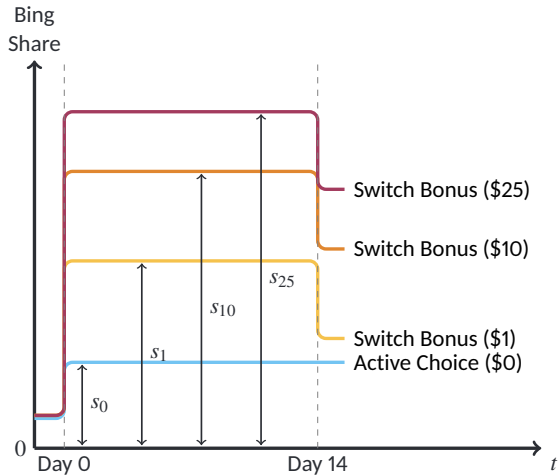


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# Identification: Idiosyncratic preferences



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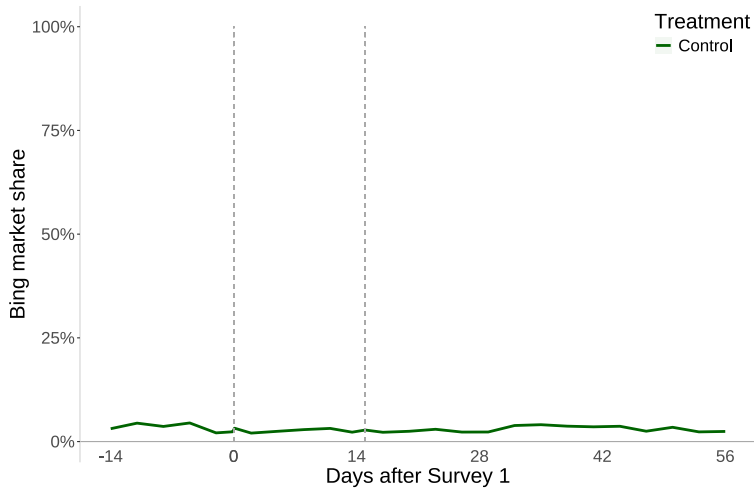


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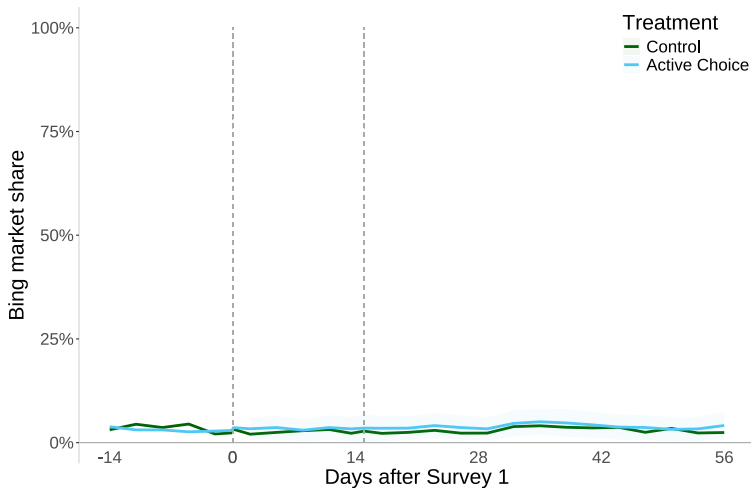


# Market shares for Chrome users



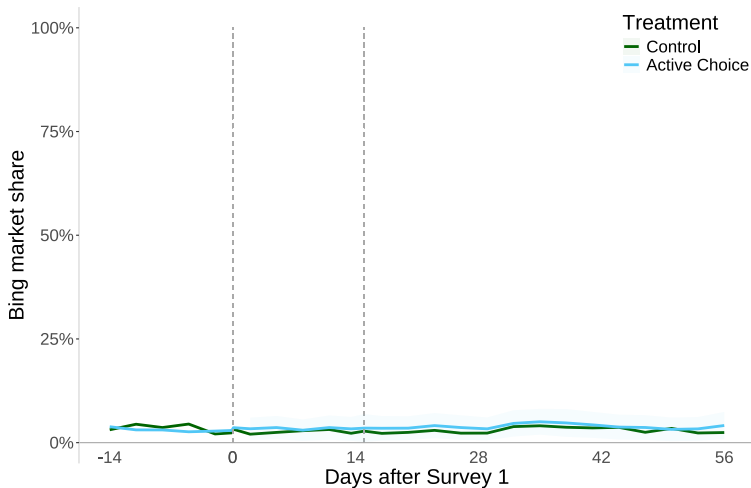
Control

# Market shares for Chrome users



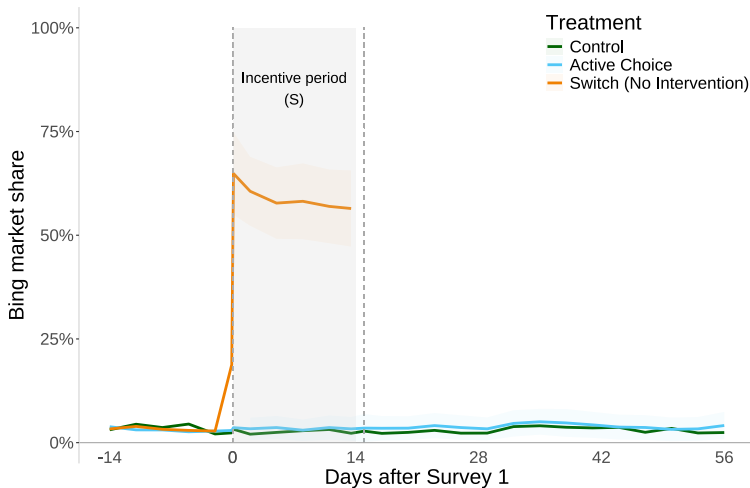
**Active Choice:** Small Bing share → Large perceived quality difference

# Market shares for Chrome users



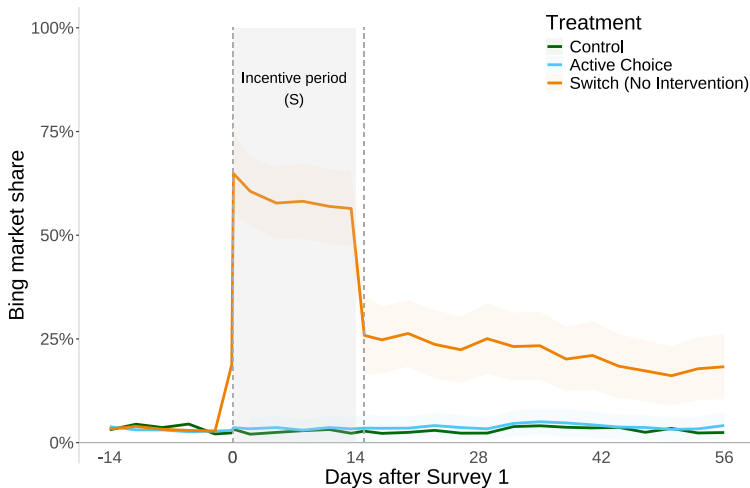
**Active Choice:** Same share as Control → Small switching cost

# Market shares for Chrome users



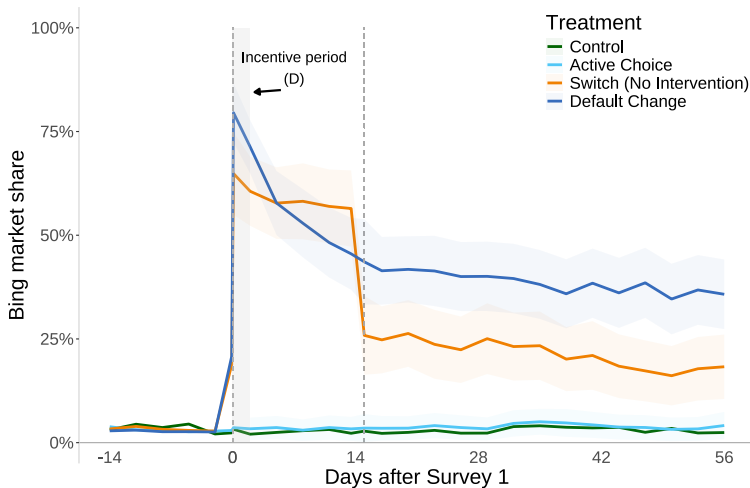
**Switch Bonus:** High Bing share during incentive → Users are price sensitive

# Market shares for Chrome users



**Switch Bonus:** Bing share stays high after incentive → Users learn about Bing

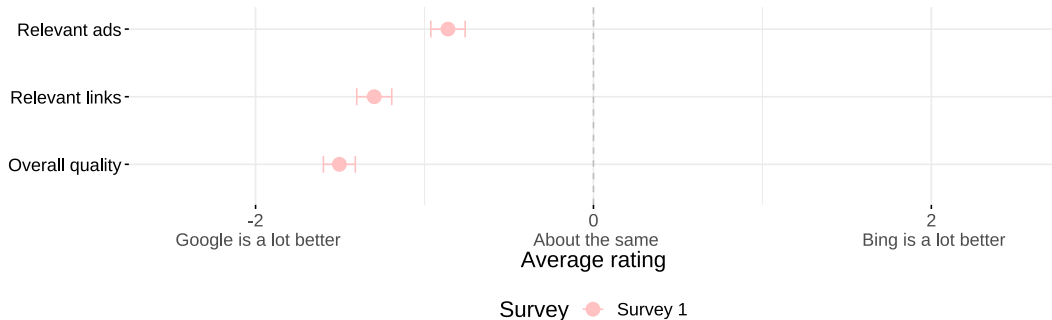
# Market shares for Chrome users



**Default Change:** Bing share converges to above Switch → There is permanent inattention

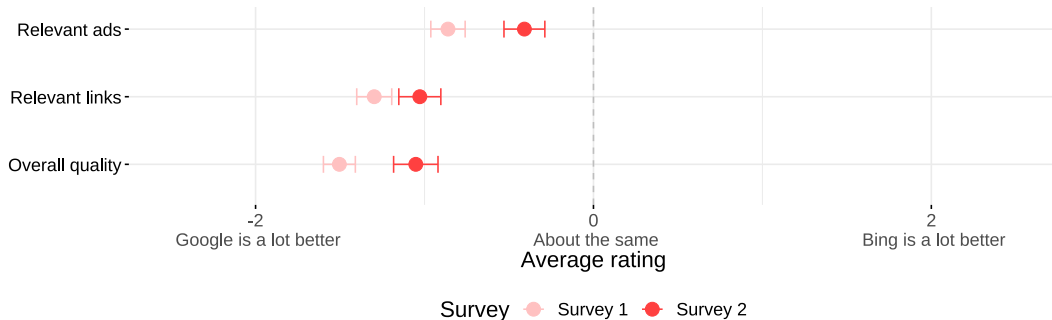
# Learning in Switch Bonus group

1. Relative preference for Bing before (Survey 1) and after (Survey 2) the switch:



# Learning in Switch Bonus group

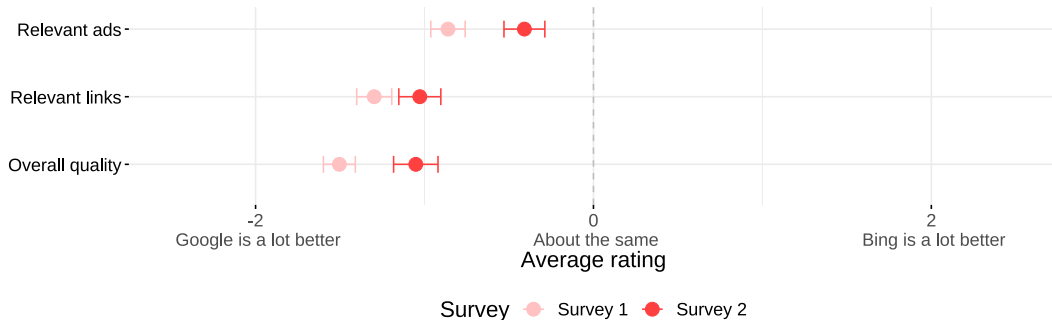
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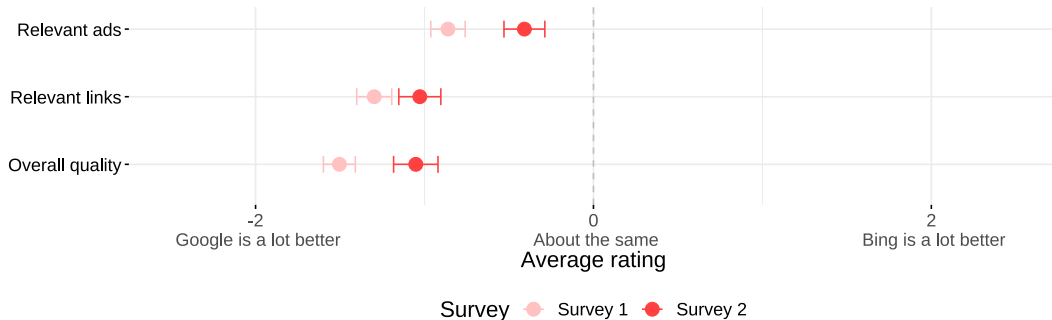


## 2. We surveyed stayers. Quotes:

- “I have learned I overall enjoy [Bing] more.”
- “I found that I liked the results I am getting in Bing”
- “I realized Bing was not as bad as I thought it was.”

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## 3. Multiple Choice:

- 65.9% – Bing better than expected
- 56.8% – They got accustomed

# Price response

## Bing market shares during incentive period:

### Active

- Bing market share  $\approx 1.86\%$

### Switch (\$1)

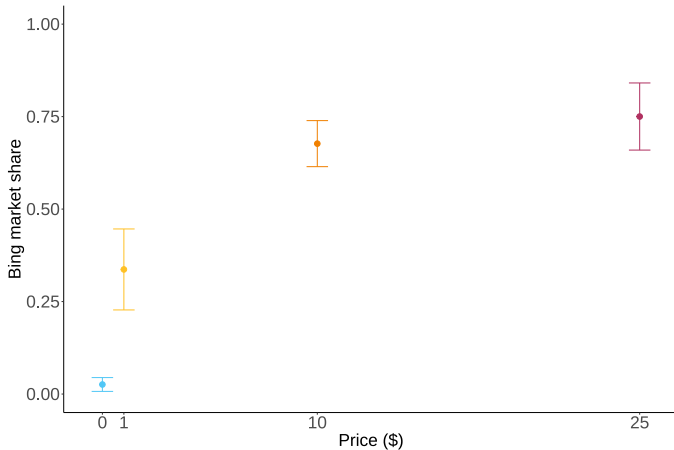
- Bing market share  $\approx 31.5\%$

### Switch (\$10)

- Bing market share  $\approx 64.4\%$

### Switch (\$25)

- Bing market share  $\approx 74.1\%$



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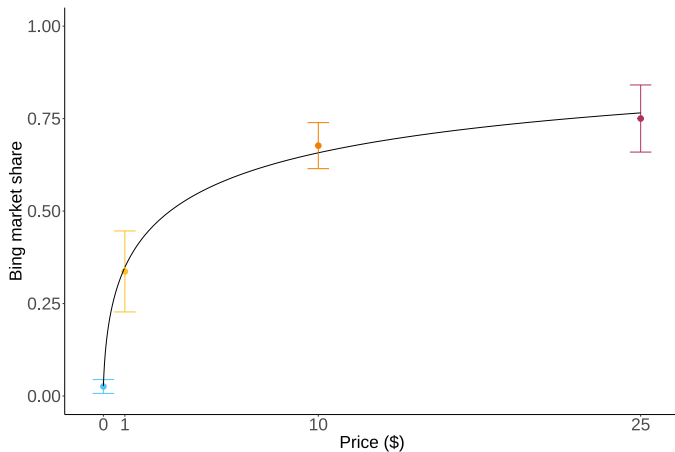
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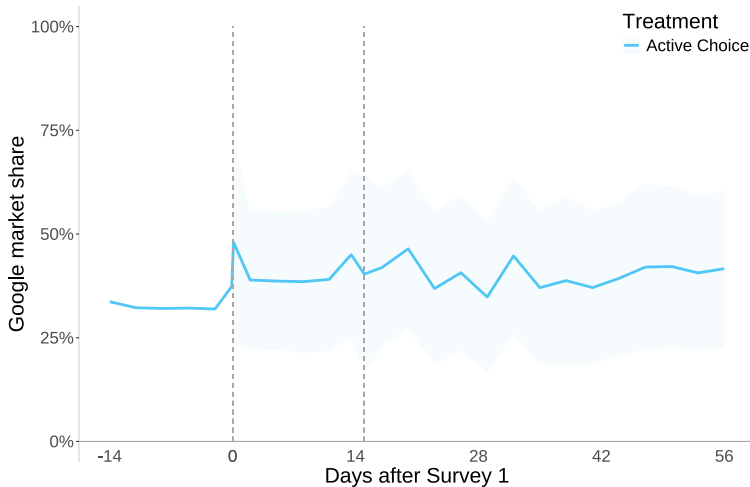
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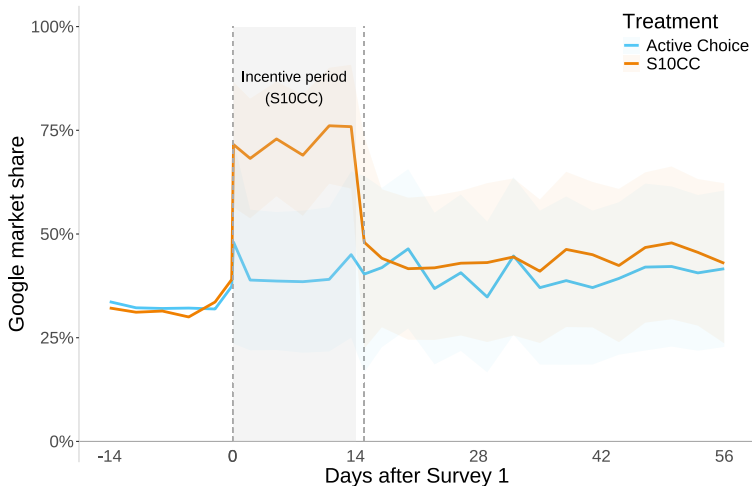


# Market shares for Edge users



**Active Choice:** Increases Google share → switching costs

## Market shares for Edge users



**\$10 Switch Bonus:** Google share matches Active Choice after incentive period → little/no learning

## Measuring users' responsiveness to result quality

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## Control (standard)

Search results for "ariel pakes linkedin" on Bing. The search bar shows "ariel pakes linkedin" and the results are categorized under "Control (standard)".

Bing found the following results

**LinkedIn**  
https://www.linkedin.com/in/ariel-pakes-5957b527

**Ariel Pakes - Professor - Harvard University | LinkedIn**  
WEB Professor. Yale University. Sep 1989 - Jan 1998 8 years 5 months. Education. Hebrew University, Jerusalem, Israel, Harvard Univeristy. Bachelor of Arts (B.A.),Masters (M.A.), ...

Tags: Ariel Pakes Harvard Harvard University

**LinkedIn**  
https://www.linkedin.com/in/ariel-pakes-898b5b11

**Ariel Pakes - Steven McArthur Heller Professor of Economics.**  
WEB View **Ariel Pakes'** profile on **LinkedIn**, the world's largest professional community. Ariel has 1 job listed on their profile. See the complete profile on **LinkedIn** and discover **Ariel's**...

Tags: Harvard University Cambridge, Massachusetts, United States Ariel Pakes

**LinkedIn**  
https://www.linkedin.com/in/ariel-pakes-315757189

**ariel Pakes - Professor - Harvard University | LinkedIn**  
WEB Professor. Harvard University. View **ariel Pakes'** profile on **LinkedIn**, the world's largest professional community. ariel has 1 job listed on their profile. See the complete profile on...

Tags: Ariel Pakes Harvard Professor at Harvard University Greater Boston

## Treatment

Search results for "ariel pakes linkedin" on Bing. The search bar shows "ariel pakes linkedin" and the results are categorized under "Treatment".

Bing found these results

**Wikipedia**  
https://en.wikipedia.org/wiki/Ariel\_Pakes

**Ariel Pakes - Wikipedia**  
WEB Ariel Stanley Pakes (born 1949) is the Thomas Professor of Economics at Harvard University. He **specializes in econometrics and industrial organization** . He is a fellow of ...

Missing: **linkedin** | Must include: **linkedin**

Tags: Harvard University Frisch Medal Ariel Pakes Steven T. Berry Born:1949

**Dunster House**  
https://dunster.harvard.edu/people/arie...

**Ariel Pakes | Dunster House**  
WEB **Ariel Pakes** is the Thomas Professor of Economics in the Department of Economics at Harvard University, where he teaches courses in Industrial Organization and Econometrics. He received the Frisch Medal of the ...

Missing: **linkedin** | Must include: **linkedin**

Tags: Harvard University Ariel Pakes Harvard Frisch Medal

**Scholars at Harvard**  
https://scholar.harvard.edu/sites/scholar.harvard...

**[PDF] Ariel Pakes - Scholars at Harvard**  
WEB Jeon J., **Pakes A.**: The Competitive Effects of Information Sharino. Finalist: Best paper



## Ranking Degradation: Effects

| Dep. var.:          | (1)                        |
|---------------------|----------------------------|
|                     | Organic click-through rate |
| Ranking Degradation | -0.077***<br>(0.028)       |
| Constant            | 0.347***<br>(0.017)        |

1. Reduced relevance of search result pages as measured by click-through rate.

## Ranking Degradation: Effects

| Dep. var.:          | (1)<br>Organic click-through rate | (2)<br>$\Delta$ Relevance rating<br>(-2 to +2 scale) |
|---------------------|-----------------------------------|------------------------------------------------------|
| Ranking Degradation | -0.077***<br>(0.028)              | -0.311***<br>(0.072)                                 |
| Constant            | 0.347***<br>(0.017)               | 0.241***<br>(0.063)                                  |

1. Reduced relevance of search result pages as measured by click-through rate.
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## Ranking Degradation: Effects

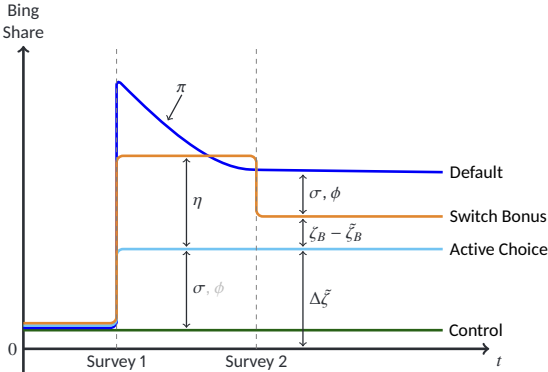
| Dep. var.:          | (1)<br>Organic click-through rate | (2)<br>$\Delta$ Relevance rating<br>(-2 to +2 scale) | (3)<br>Bing share   |
|---------------------|-----------------------------------|------------------------------------------------------|---------------------|
| Ranking Degradation | -0.077***<br>(0.028)              | -0.311***<br>(0.072)                                 | -0.034<br>(0.029)   |
| Constant            | 0.347***<br>(0.017)               | 0.241***<br>(0.063)                                  | 0.244***<br>(0.027) |

1. Reduced relevance of search result pages as measured by click-through rate.
2. Worsened participants' perception of result relevance.
3. Had only limited impact on participant choices.

# Demand estimation results

| Description           | Formula                       | Estimate | 95% CI         |
|-----------------------|-------------------------------|----------|----------------|
| Permanent inattention | $\phi$                        | 0.34     | [0.20,0.41]    |
| Attention probability | $\pi$                         | 0.82     | [0.38,0.97]    |
| Price response        | $\eta$                        | 0.31     | [0.22,0.55]    |
| Switching cost        | $\sigma$                      | \$0.004  | [-0.005,0.034] |
| Perceived Bing pref.  | $\Delta \tilde{\zeta}$        | -\$3.20  | [-4.49,-1.85]  |
| Learning              | $\zeta_B^* - \tilde{\zeta}_B$ | \$0.21   | [0.04,0.58]    |
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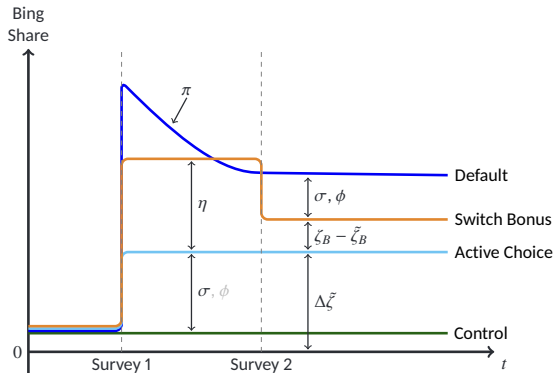
For presentation, Chrome users only.



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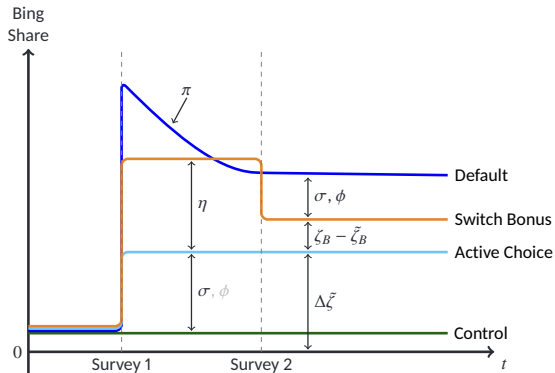


- 34% of users are permanently inattentive

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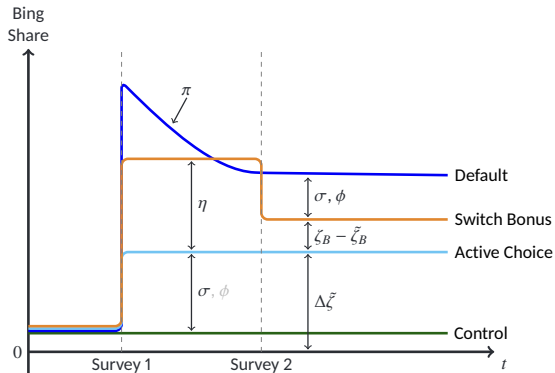


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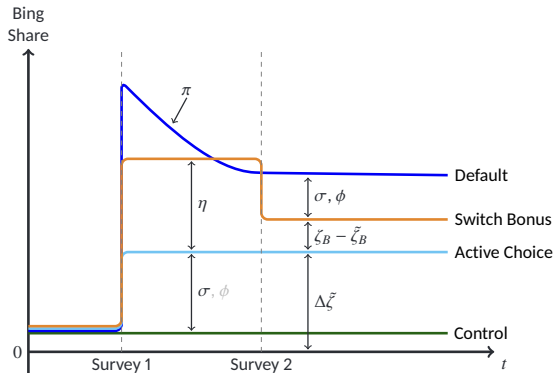


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- 34% of users are permanently inattentive
- If users make active choice, Bing payment of \$3.20 per two weeks equalizes market shares
- If perceptions were corrected, required payment would shrink to \$2.99
- Limited demand response to result relevance



# Agenda

1. Demand Model
2. Experimental Design
3. Identification
4. Experimental Results and Estimation
- 5. Data Scale Economies**
6. Counterfactuals

# Data Scale Economies: Overview

More users (and data) → Better search results → More users

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- Random sample of 43,991 queries that are new, i.e.:
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  - $\geq 100$  searches in sample year (2022)
- For each impression of each query we observe
  - Timestamp
  - (Anonymized) identifier for top result link shown
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- For each impression of each query we observe
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  - (Anonymized) identifier for top result link shown
  - Whether the user clicked on the top result

**Approach:**

1. Estimate how any given query's click-through rate (CTR) increases with # of impressions
2. Integrate over query frequency distribution (probably effect concentrates on long tail)

## Data Scale Economies: Identification

- Challenge: popularity & ease are correlated (e.g. “facebook”, “amazon”)  
→ Use within-query variation over time

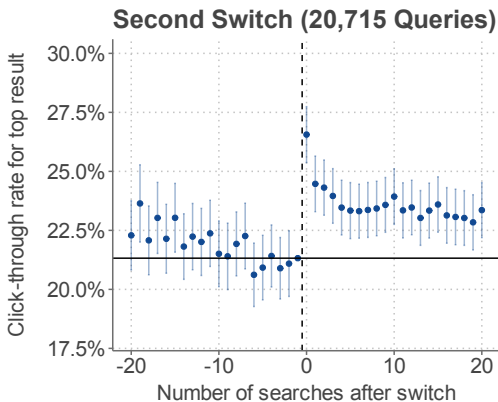
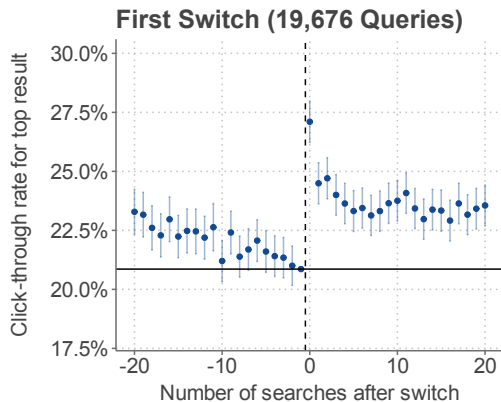


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## Data Scale Economies: Model & Estimation

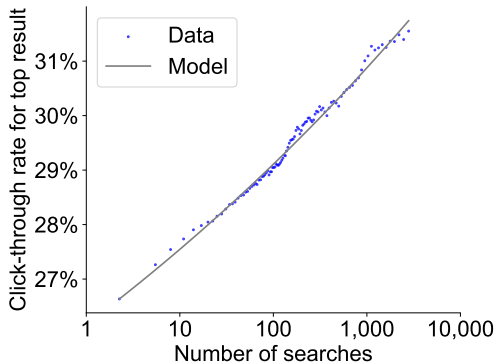
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# Data Scale Economies: Model & Estimation

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| Description      | Parameter          | Estimate | SE   |
|------------------|--------------------|----------|------|
| CTR at inception | $\alpha$           | 0.18     | -    |
| Value of data    | $100 \times \beta$ | 0.56     | 0.07 |
| Shape of returns | $\theta$           | 0.95     | 0.03 |

→ returns to scale are logarithmic

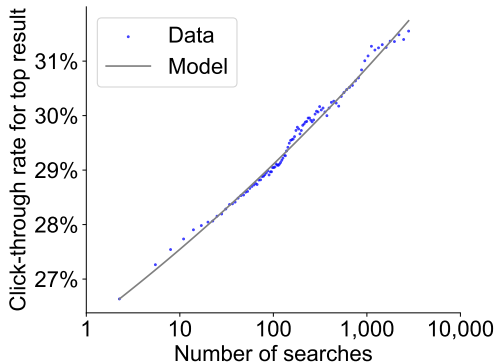


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- If Bing had access to Google's data, CTR would increase from 23.5% to 24.8%.
- Caveats: observational data, estimated only off 'new' search terms, no cross-query learning.

# Agenda

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# Counterfactuals

|             |                                  |                      | Direct effects<br>(fixed quality) |                      |
|-------------|----------------------------------|----------------------|-----------------------------------|----------------------|
| Description | Switching cost<br>& inattention? | Misper-<br>ceptions? | Google<br>share (%)               | CS gain<br>(\$/year) |
| Status Quo  | ✓                                | ✓                    | 88.9                              | 0.00                 |

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| Status Quo   | ✓                                | ✓                    | 88.9                              | 0.00                 |
| No Frictions |                                  |                      | 73.8                              | 6.01                 |

Eliminating demand-side frictions reduces Google market share (with moderate CS gain).



## Counterfactuals

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| No Frictions         |                                  |                      | 73.8                              | 6.01                 |
| Chrome Choice Screen |                                  | ✓                    | 87.6                              | 0.09                 |

An active choice screen leaves shares unchanged, but gets most CS gains.

# Counterfactuals

| Description          | Switching cost<br>& inattention? | Misper-<br>ceptions? | Direct effects<br>(fixed quality) |                      |
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| No Frictions         |                                  |                      | 73.8                              | 6.01                 |
| Chrome Choice Screen |                                  | ✓                    | 87.6                              | 0.09                 |
| No Misperceptions    | ✓                                |                      | 78.4                              | 0.46                 |

Correcting perceptions lowers Google share, but with small CS change.

# Counterfactuals

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| No Frictions           |                                  |                      | 73.8                              | 6.01                 |
| Chrome Choice Screen   |                                  | ✓                    | 87.6                              | 0.09                 |
| No Misperceptions      | ✓                                |                      | 78.4                              | 0.46                 |
| Bing Default on Chrome | ✓                                | ✓                    | 48.9                              | -70.92               |

Making Bing the default lowers Google share, but at a large CS loss.

# Counterfactuals

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| No Frictions                 |                                  |                      | 73.8                              | 6.01                 |
| Chrome Choice Screen         |                                  | ✓                    | 87.6                              | 0.09                 |
| No Misperceptions            | ✓                                |                      | 78.4                              | 0.46                 |
| Bing Default on Chrome       | ✓                                | ✓                    | 48.9                              | -70.92               |
| + Delayed Chrome Choice Scr. | ✓                                | ✓                    | 72.1                              | 0.04                 |

Delayed choice screen: shows up two weeks after browser installation

→ Reduces Google share at a small CS loss

# Counterfactuals

| Description                  |   |   | Direct effects<br>(fixed quality) |                      | Equilibrium effects<br>(endogenous quality) |                      |
|------------------------------|---|---|-----------------------------------|----------------------|---------------------------------------------|----------------------|
|                              |   |   | Google<br>share (%)               | CS gain<br>(\$/year) | Google<br>share (%)                         | CS gain<br>(\$/year) |
| Status Quo                   | ✓ | ✓ | 88.9                              | 0.00                 |                                             |                      |
| No Frictions                 |   |   | 73.8                              | 6.01                 |                                             |                      |
| Chrome Choice Screen         |   | ✓ | 87.6                              | 0.09                 |                                             |                      |
| No Misperceptions            | ✓ |   | 78.4                              | 0.46                 |                                             |                      |
| Bing Default on Chrome       | ✓ | ✓ | 48.9                              | -70.92               |                                             |                      |
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| Status Quo                   | ✓                                | ✓                    | 88.9                              | 0.00                 | 88.9                                        | 0.00                 |
| No Frictions                 |                                  |                      | 73.8                              | 6.01                 | 73.5                                        | 6.02                 |
| Chrome Choice Screen         |                                  | ✓                    | 87.6                              | 0.09                 | 87.6                                        | 0.09                 |
| No Misperceptions            | ✓                                |                      | 78.4                              | 0.46                 | 78.2                                        | 0.47                 |
| Bing Default on Chrome       | ✓                                | ✓                    | 48.9                              | -70.92               | 48.5                                        | -70.81               |
| + Delayed Chrome Choice Scr. | ✓                                | ✓                    | 72.1                              | 0.04                 | 72.0                                        | 0.05                 |

Data feedback has only minor effects

→ Small demand response to result relevance + small effect of data on result relevance

# Counterfactuals

| Description                  | Switching cost<br>& inattention? | Misper-<br>ceptions? | Direct effects<br>(fixed quality) |                      | Equilibrium effects<br>(endogenous quality) |                      |
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| Status Quo                   | ✓                                | ✓                    | 88.9                              | 0.00                 | 88.9                                        | 0.00                 |
| No Frictions                 |                                  |                      | 73.8                              | 6.01                 | 73.5                                        | 6.02                 |
| + Data Sharing               |                                  |                      |                                   |                      | 73.1                                        | 6.12                 |
| Chrome Choice Screen         |                                  | ✓                    | 87.6                              | 0.09                 | 87.6                                        | 0.09                 |
| No Misperceptions            | ✓                                |                      | 78.4                              | 0.46                 | 78.2                                        | 0.47                 |
| + Data Sharing               |                                  |                      |                                   |                      | 77.9                                        | 0.56                 |
| Bing Default on Chrome       | ✓                                | ✓                    | 48.9                              | -70.92               | 48.5                                        | -70.81               |
| + Delayed Chrome Choice Scr. | ✓                                | ✓                    | 72.1                              | 0.04                 | 72.0                                        | 0.05                 |

Data sharing only has minor effects

# Counterfactuals

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| No Frictions                 |                                  |                      | 73.8                              | 6.01                 | 72.9                                        | 6.04                 |
| + Data Sharing               |                                  |                      |                                   |                      | 71.9                                        | 6.31                 |
| Chrome Choice Screen         |                                  | ✓                    | 87.6                              | 0.09                 | 87.5                                        | 0.09                 |
| No Misperceptions            | ✓                                |                      | 78.4                              | 0.46                 | 77.9                                        | 0.47                 |
| + Data Sharing               |                                  |                      |                                   |                      | 77.1                                        | 0.72                 |
| Bing Default on Chrome       | ✓                                | ✓                    | 48.9                              | -70.92               | 47.7                                        | -70.61               |
| + Delayed Chrome Choice Scr. | ✓                                | ✓                    | 72.1                              | 0.04                 | 71.4                                        | 0.08                 |

Also minor effects at the 95% CI boundary of our estimate of demand response to quality



# Conclusion

## Defaults are effective

- $\approx 1/3$  of users are permanently inattentive
- Prevent users from learning about other search engines

Part of Google's advantage lies in **consumers misperceiving** the quality of rivals

How can regulators **reduce Google's market share?**

- Choice screens alone do not move the needle (Decarolis, Li, and Paternello; 2023)
- Changing the default does, but w/ large decrease in CS
- Successful policies need to account for the experience goods aspect

**Economies of scale** and data sharing have small effects

# Google vs DOJ: Remedy Decision

## 1. Chrome choice screen: ✗

- "(...) choice screens have not been shown to enhance competition"
- "Precedent (...) avoid remedies that compel product design requirements"

## 2. Banning payments for default: ✗

- "(...) substantial—in some cases, crippling—downstream harms to distribution partners, related markets, and consumers." (e.g., Apple, Samsung, Mozilla)
- "Freed of having to pay billions in revenue share, Google's profits would *increase*."

## 3. Data sharing: ✓

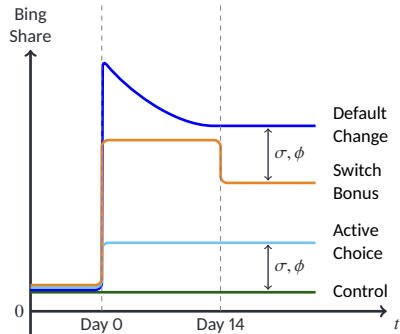
- "(...) will deny Google the fruits of its exclusionary acts and promote competition"

## 4. Ban on exclusive contracts: ✓

## 5. Divesting Chrome and Android: ✗

**Thank you!**

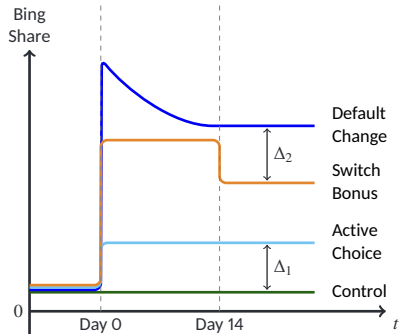
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**Key idea:** Switching costs and permanent inattention affect  $\Delta_1$  and  $\Delta_2$  quite differently

- A and S unaffected by inertia  $\implies$  focus on effects on C and D



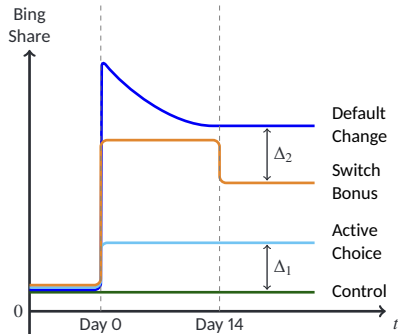
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## Effect of permanent inattention $\phi$ :

- In C, only affects people who would like to overrule default and use Bing ( $< 5\%$ )
- In D, affects all users who accepted payment but would want to switch back ( $\sim 50\%$ )
- Effect on  $\Delta_2$  much larger than on  $\Delta_1$ .



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- In D, affects all users who accepted payment but would want to switch back ( $\sim 50\%$ )
- Effect on  $\Delta_2$  much larger than on  $\Delta_1$ .

## Effect of switching cost $\sigma$ :

- $\sigma$  shifts utilities in C and D by the same amount
- Affect  $\Delta_1$  and  $\Delta_2$  approx. symmetrically
  - as long as similar densities of marginal users (relies on identifying error distribution)

